

Vix Talbot

Data Scientist & MSAI Candidate, Johns Hopkins University

I'm a data scientist with a background in consulting, now pursuing my Master's in Artificial Intelligence at Johns Hopkins as I transition toward machine learning engineering. My work spans computational stylometry, healthcare research, and NLP.

Integrity and rigor guide how I work. During a neuroscience internship, my final presentation included data I'd carefully processed; a senior researcher in attendance recognized an inconsistency it revealed, which traced back to a flaw in the study's underlying protocol. Watching that come to light shaped how I think about transparency and reproducibility. I care as much about how a model is built as what it predicts, and about bringing the people I collaborate with along with genuine care, not just technical competence.

In my freetime I enjoy crocheting, SUP boarding in the beautiful PNW, and contra dancing.



Contact

Email: talbot.vix@gmail.com
LinkedIn: [linkedin.com/in/vix-talbot](https://www.linkedin.com/in/vix-talbot)

Home

Contact

Projects

Publications

Projects

Personal Website (Module 1 - Flask and Blueprints)

This site is a three-page personal portfolio built with Flask, structured around the application factory pattern with Blueprints to keep route logic modular and organized. It features a persistent, top-right navigation bar with active-page highlighting driven by Flask's request object, externally managed CSS for clean separation of structure and style, and a fully reproducible environment captured in requirements.txt.

[View on GitHub](#)

RELATED WORK

Complex Network Analysis of Russian Translations

To address the challenge in translated literature research, where a translator's individual style is obscured by the original author's semantic content, this project applies complex network analysis alongside classical stylometry to help define the subtle styles of five Russian translations of Mark Twain's *The Adventures of Huckleberry Finn* (1911-1960). The computational analysis revealed statistically significant divergences in both lexical and

discovered topological anomalies: the 1960 Daruzes translation uniquely inverts the standard grammatical hierarchy by prioritizing verbs over nouns. Finally, betweenness centrality analysis found a wide range of data when ranking the most connective nodes across the corpus, suggesting that different translators had quite different preferences for which stop words they relied upon.

[View on GitHub](#)

Stylometric Analysis of Mark Twain's The Adventures of Huckleberry Finn in Russian Translation

Dr. Cassio de Oliveira, a professor of Russian literature at Portland State University, was writing a book on Mark Twain's reception in the Soviet Union and needed quantitative support for his qualitative analysis of five Russian translations spanning 1911-1960. As lead of a five-person consulting team through PSU's Computation and Data-Enabled Science NSF Research Training Group, I directed the technical execution of a stylometric analysis using Burrows' Delta method, building an R-based NLP pipeline (udpipe, tm, proxy, factoextra) to lemmatize, vectorize, and compare the five translations by word frequency, part-of-speech patterns, and lexical diversity. The pipeline used PCA for dimensionality reduction and cosine similarity to quantify stylistic distance between every pair of translations, producing similarity matrices and cluster visualizations Dr. de Oliveira could cite directly in his book. Beyond the technical build, I led team workflow: since our program hadn't covered version control, I brought in an outside Git consultant to train the team on collaborative coding practices, and ensured each member understood the analysis deeply enough to present it confidently to the client.

[View on GitHub](#)

[Home](#)

[Contact](#)

[Projects](#)

[Publications](#)

Publications

Examining the Interaction Between Calcium Supplement Use, Demographics, and Lifestyle Factors in Postmenopausal Bone Health

Osteoporosis is a condition which poses a significant health threat, particularly among women during the menopause transition, where accelerated bone loss increases fracture risk. Calcium supplementation has been shown to be an important intervention to mitigate bone mineral density (BMD) decline during this and other periods of life. However, the efficacy of calcium supplementation is influenced by various individual factors, including demographics and lifestyle habits. This study investigates the interaction between calcium supplement use, and several interaction terms on bone health in women. Multiple linear regression analysis is employed to assess the impact of these factors on BMD. Data from the Study of Women's Health Across the Nation (SWAN) dataset, representing diverse racial/ethnic backgrounds, are utilized. The findings aim to inform personalized strategies for optimizing bone health and preventing osteoporotic fractures in postmenopausal women.

[Read the thesis](#)